Atomic elementary flux modes explain the steady state flow of metabolites in large-scale flux networks

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Abstract

Steady state fluxes are a measure of cellular activity under metabolic homoeostasis, but understanding how individual substrates are metabolised remains a challenge in large-scale networks. Pathway-based approaches such as elementary flux mode (EFM) analysis are limited to small networks due to the combinatorial explosion of pathways and the ambiguity of decomposing fluxes onto EFMs. Here, we present an alternative approach to explain metabolic fluxes in terms of the steady state flow of their atomic constituents. We refer to these pathways as atomic elementary flux modes (AEFMs) and show that computations involving AEFMs are orders of magnitude faster than standard EFMs. Using our approach, we enumerate carbon and nitrogen AEFMs in five genome-scale metabolic models and compute the AEFM decomposition of fluxes estimated in a HepG2 liver cancer cell line. Our results systematically characterise carbon and nitrogen remodelling and, on the HepG2 network, predict glutamine metabolism through a recently discovered non-canonical TCA cycle.

Introduction

Recent technological and computational advancements are leading to a rise in metabolic flux data (Stern et al, 2023; Lee et al, 2019; Wang et al, 2022a,b). These fluxes quantify the rate of metabolite interconversions within a cell or their movements across subcellular compartments. Metabolic fluxes are studied to understand the dynamics of substrate switching (Zampieri et al, 2019; Kochanowski et al, 2021), redox balancing (McKinlay & Harwood, 2010; Franchina et al, 2022), overflow metabolism (Basan et al, 2015; Vander Heiden et al, 2009), cell differentiation (Endo et al, 2022; Jackson & Finley, 2024), and cellular communities (Schuetz et al, 2012; Yu et al, 2022; Gustafsson et al, 2024). Hence, 'fluxomics' (Emwas et al, 2022; Sanford et al, 2002) is emerging as a field dedicated to quantifying the totality of fluxes within biological systems.

Current experimental and computational methods generally estimate fluxes under metabolic homeostasis. Under this assumption, the production and consumption of metabolites are balanced over time, and the fluxes are said to be under steady state. Techniques such stable isotope tracing analysis (SITA) experimentally determine steady state fluxes by culturing cells in labelled substrates and quantifying their incorporation within endogenous metabolites (Buescher et al, 2015; Antoniewicz, 2018). Using this method, fluxes have been measured in major metabolic subsystems such as glycolysis, pentose phosphate pathway, and the TCA cycle (Zamboni et al, 2009; Ahn et al, 2017; Wu et al, 2019). Many computational methods, including but not limited to flux balance analysis (FBA), have also been proposed to infer steady state fluxes from transcript, protein, and metabolite abundance data (Uematsu et al, 2022; González-Arrué et al, 2023; Huang et al, 2023; Kaste & Shachar-Hill, 2023; Lee

et al, 2023). Collectively, these approaches are being used to estimate and predict fluxes in genome-scale metabolic models (GEMs) involving hundreds to thousands of reactions (Nilsson et al, 2020b; Cherkaoui et al, 2022; Gelbach et al, 2024).

Although much effort has been made to estimate steady state fluxes, less emphasis has been placed on developing methods to analyze these large-scale flux datasets. In contrast to abundance data (e.g. transcripts, proteins, metabolites), steady state fluxes are constrained by the reaction stoichiometries between metabolite inflows and outflows. In a network with fixed source and sink fluxes, an increase in flux along one metabolic pathway must decrease fluxes along one or more other pathways to maintain mass conservation. For simple networks, one may predict how the system would evolve by identifying the up and downstream reactions flanking an enzymatic perturbation. For example, Schwartz & Kanehisa were able to enumerate all glycolytic pathway fluxes in a 12-reaction model of yeast glycolysis (Schwartz & Kanehisa, 2006). Yet in more complex genome-scale metabolic networks containing thousands of fluxes (e.g. Brunk et al (2018); Robinson et al (2020); Wang et al (2021)), there exists no hierarchical relationship between reactions. A change in one reaction rate, let alone multiple perturbations, could alter network fluxes well beyond neighbouring reactions (Nobile et al, 2021).

This problem of understanding the flow of metabolites within steady state flux networks has historically been addressed through the concept of elementary flux modes (EFMs). EFMs are a pathway-based approach to analyze metabolic networks and are defined as minimal sets of biochemical reactions that maintain steady state flux (Schuster & Hilgetag, 1994). This minimal property specifies that an EFM cannot be decomposed into two or more smaller pathways carrying steady state flux. An attractive property of EFMs is that any set of steady

state fluxes in a network can be explained as a positive, linear combination of EFMs (Schuster et al, 1999). While EFMs are a mechanistic explanation of how metabolites travel through networks, the number of EFMs scales combinatorially as a function of network size (Acuña et al, 2009). It thus remains computationally infeasible to enumerate EFMs in GEMs after over a decade of algorithmic advancements (e.g. Gagneur & Klamt 2004; Terzer & Stelling 2008; Buchner & Zanghellini 2021). The number of EFMs also poses a challenge for explaining network fluxes in terms of their EFMs. When there are more EFMs than linearly independent fluxes, there exist multiple ways to assign EFM weights that reconstruct the observed network fluxes. Consequently, structural analyses of EFMs and downstream analyses of their weights have been limited to small-scale metabolic subnetworks (Fakih et al, 2023; Mpabanzi et al, 2022; Toya & Shimizu, 2022; Matsuda et al, 2011; Schwartz & Kanehisa, 2006).

In parallel with EFM analysis, the task of identifying metabolic pathways has also been approached by numerous other graph-theoretic approaches. These methods derive graphs based, in part, on the metabolic flux network (e.g. Figure 1a) that highlight specific relationships between substrates and products. The simplest representation, for example, is the metabolic graph (Arita, 2000; Wagner & Fell, 2001) (Figure 1b), whereby metabolite nodes are connected to each other based on substrate-product reaction relationships. These graphs are typically analysed by identifying the K-shortest pathways between source and target metabolites. However, these methods often returned biologically meaningless pathways due to metabolite byproducts and co-factors, leading to pathways that inaccurately described the remodelling of source/target metabolite pairs. Subsequent refinements implemented heuristics to avoid enumerating these pathways (McShan et al, 2003; Rahman et al, 2004; Croes

et al, 2006; Pey et al, 2011a; Hafner & Hatzimanikatis, 2021); however, this problem was fully solved by the introduction of atomic transition graphs (Arita, 2003). These graphs use atom mapping data to model the movement of individual atoms across substrates and products, thereby avoiding metabolic byproducts (Figure 1c). We refer to weighted versions of these graphs as *atomic flux graphs* which are used in isotope-based metabolic flux analysis to predict intracellular reaction fluxes (Wiechert et al, 1999; Weitzel et al, 2007).

Current methods to analyse metabolic flux networks and atomic flux graphs are often viewed as independent approaches for metabolic pathway analysis (de Figueiredo et al, 2009b; Algfoor et al, 2017). However, both can be viewed as related types of flux networks operating at the metabolite or metabolite-atom level. One could envision applying metabolic pathway analysis techniques designed for metabolic flux networks on their corresponding atomic flux graphs. This ideas was previously explored by Pey et al (Pey et al, 2011b) who proposed the concept of elementary carbon modes (ECMs). These ECMs were defined as minimal, mass-balanced pathways tracing the flow of individual carbon atoms in an atomic flux graph (Pey et al, 2011b, 2012). Like paths in metabolite-atom graphs, ECMs correspond to either simple paths spanning source and sink metabolite carbons or simple cycles involving internal metabolite-carbon positions.

Since ECMs are specific to carbon pathways, we propose generalising ECMs to any element in an atomic flux network. We refer to these pathways as *atomic elementary flux modes* (AEFMs) and formally define them as the EFMs of an atomic flux graph. To avoid confusion with classical EFMs of metabolic flux networks, we henceforth refer to these pathways as *molecular* EFMs to emphasize their balance of molecular stoichiometries. Figure 1d shows the molecular EFMs for the metabolic flux network in Figure 1a, while Figure 1e shows the

AEFMs for the corresponding atomic transition/flux graph. Although there are a different number of molecular EFMs-versus-AEFMs, both pathways carry some form of steady state flux. The molecular EFMs balance the reaction fluxes and molecular stoichiometries, while the AEFMs balance the atomic fluxes and atom mass transfers. We further observe that the flux through each reaction is equal to the underlying fluxes at the atomic level (Pey et al, 2012). This leads to the following linear relationship between molecular EFM and AEFM weights: $w_1 = x_1 = x_2$, and $w_2 = x_3 = x_4 = y_1$. Thus in Figure 1d-e, knowledge of either x_1/x_2 and $x_3/x_4/y_1$ in the atomic flux graph is sufficient to determine the EFM weights in the original metabolic flux network.

Although corresponding AEFM and molecular EFM weights presents a new computational challenge, we focus our attention on two fundamental problems associated with AEFMs. Following molecular EFMs, these problems are enumerating and uniquely decomposing atomic fluxes onto AEFMs. Addressing the latter problem is relevant since the number of AEFMs can exceed the number of linearly independent fluxes in more complex atomic flux graphs. Indeed, Pey et al encountered this latter problem with ECMs and were unable to uniquely estimate their weights beyond simple carbon flux graphs (Pey et al, 2012).

In our previous work, we developed a novel solution to the EFM flux decomposition problem for the special case of unimolecular reaction networks under a Markov constraint. We modelled a hypothetical particle moving between metabolites under the assumption that the transition probability of each reaction was proportional to the observed steady state flux. By expanding the notion of Markov chain state to include sequences of metabolites, we proposed an algorithm to efficiently compute simple cycle probabilities corresponding to a given EFM. We called this construct the *cycle-history Markov chain* (CHMC) and proved

that these EFM probabilities could be scaled to weights that fully reconstructed the flux along each reaction (Chitpin & Perkins, 2023).

Here, we combine three ideas in metabolic modelling to produce a pipeline for analysing metabolic flux networks under steady state. First, we use the state-of-the art atom mapping program RXNMapper (assisted by manual curation) to automatically construct atomic flux graphs for any given metabolic network. Second, we merge the concept of metabolite-atom paths with elementary flux modes through our proposal of AEFMs. Third, we expand the CHMC method to operate on atomic transition/flux graphs to enumerate and compute unique AEFM weights. Using this pipeline, we demonstrate the feasibility of AEFM enumeration by showing that the number of these carbon and nitrogen pathways are orders of magnitude smaller than the number of molecular EFMs in five metabolic networks. We then analyse the structure of these AEFMs characterising carbon and nitrogen remodelling within and across nutrient sources. Lastly, we run our full pipeline on a HepG2 metabolic flux network to examine glutamine carbon remodelling. Our AEFM weight analysis reveals that glutamine consumption is routed through a small number of highly active AEFMs, including a recently discovered non-canonical TCA cycle (Arnold et al, 2022).

Results

Pipeline for AEFM enumeration and weight identification

We first describe what molecular EFMs look like in biologically-relevant metabolic networks and provide intuition why enumerating and decomposing fluxes onto these pathways

remains a fundamental problem. Figure 2a shows an example molecular EFM involving substrates glucose and ATP which are consumed to produce metabolites G3P, DHAP, and ADP. Multiple source and sink fluxes are required to balance the internal metabolite stoichiometries, leading to several branching and converging paths within the molecular EFM. As the metabolic networks expands to include dozens or more reactions, the corresponding molecular EFMs can grow increasingly complex with numerous source and sink fluxes required to stoichiometrically balance participating metabolites. In particular, there may be multiple sets of reactions involved to balance pairs of cofactors (e.g. ATP/ADP, NAD^+/NADH), metabolic byproducts (e.g. water, hydrogen ions), or coenzymes (e.g. coenzyme A). This complexity can lead to an intractable number of pathways that may only differ by reaction subsets required to mass-balance these ubiquitous metabolites. Removing these metabolites may, in part, solve this problem; however, the choice of metabolite to prune is often subjective and not guaranteed to sufficiently reduce the total number of molecular EFMs.

Given many metabolic flux networks are generated to study nutrient catabolism (Zamboni et al, 2015) or engineer desirable metabolic products (Zanghellini et al, 2013), we argue for studying source metabolite-derived AEFMs to reduce the combinatorial number of molecular EFMs. We highlight an example carbon AEFM in Figure 2b which shows how the C6 glucose carbon is remodelled during the first steps of glycolysis. As atoms are never split apart or fused together in metabolic reactions, these AEFMs always correspond to simple paths spanning source and sink metabolites or simple cycles between internal metabolites. These pathways are straightforward to interpret because each atom within an AEFM can only transit from a single substrate to product, regardless of the network size, connectivity, or number of participating substrates/products in each reaction. Hence, this property enables

us to enumerate and compute the AEFM weights using our previously described CHMC method to uniquely decompose steady state fluxes onto EFMs for unimolecular reaction networks (Chitpin & Perkins, 2023).

We outline our computational pipeline in Figure 2c to enumerate AEFMs and uniquely assign their weights when the steady state network fluxes are known. In step 1, we require as inputs the metabolic network defined by a set of metabolites and reactions encoded as a stoichiometry matrix, the metabolite structures denoted as SMILES strings, and the steady state fluxes. From the stoichiometry matrix and SMILES strings, we construct reaction SMILES strings which are imputed into the atom mapping program RXNMapper (Schwaller et al, 2021) to map the position of each atom within each reaction equation.

Using these atom mapping predictions, we construct atomic transition/flux subgraphs rooted on each source metabolite atom entering the metabolic network. Figure 2c shows an example atomic transition/flux graph for the C6 carbon of input glucose entering the network with atomic fluxes proportional to the molecular fluxes, accounting for reactions with multiple substrate stoichiometries (see Methods and Figures S1-S2). In step 3, we construct the corresponding ACHMC to enumerate the AEFMs for each atomic transition/flux graph. Finally, the AEFM weights are computed in step 4 by steady state analysis of the ACHMC.

Enumerating AEFMs is computationally tractable in large-scale networks

Given the major computational challenge of enumerating molecular EFMs in networks with > 100 metabolites and reactions, we first sought to test whether it was feasible to enumer-

ate AEFMs in GEMs. We selected five GEMs across distinct organisms in ascending order of network size. These networks include a metabolic model of *E. coli* (E. coli core; Orth et al 2010b; Data ref: Orth et al 2010a), human red blood cells (iAB RBC 283; Bordbar et al 2011b; Data ref: Bordbar et al 2011a), *H. pylori* (iIT341; Thiele et al 2005b; Data ref: Thiele et al 2005a), *S. aureus* (iSB619; Becker & Palsson 2005b; Data ref: Becker & Palsson 2005a), and human liver cancer cells (HepG2; Nilsson et al 2020b; Data ref: Nilsson et al 2020a). These models were pre-processed to ensure unambiguous atom mappings across all reactions. Namely, all metabolites with no known structures (pseudometabolites), and pseudoreactions with non-integer stoichiometries were removed from the networks. The resulting metabolic networks ranged between 71-547 metabolites and 144-974 reactions (Figure S3). We then attempted to enumerate the molecular EFMs using FluxModeCalculator, which is the fastest molecular EFM enumeration program (van Klinken & Willems van Dijk, 2016), and our AEFM enumeration pipeline (Figure 1c steps 1-3). We first compiled the metabolite SMILES strings and constructed ACHMCs rooted on each carbon and nitrogen atom across all source metabolites. The resulting summary statistics describing the GEMs and ACHMCs are presented in Table 1. Across all carbon ACHMCs, we observed a broad increase in the total number of AEFMs as a function of metabolic network size. The one exception to this observation was the HepG2 network which contained fewer metabolites and reactions than iSB619, yet exhibited two orders of magnitude more carbon AEFMs. Inspection of the HepG2 network revealed it only contained 40 source metabolites compared to the 198 source metabolites in the iSB619 network, indicating a greater degree of network connectivity between internal metabolites (Figure S3).

Figure 3a shows the total number of AEFMs and molecular EFMs enumerated by our

program MarkovWeightedEFMs.jl and FluxModeCalculator on the five GEMs. Using our ACHMC pipeline, we enumerated between $10^3 - 10^6$ carbon and $10^2 - 10^4$ nitrogen AEFMs within each GEM. In contrast, FluxModeCalculator failed to enumerate molecular EFMs in all but the two smallest E. coli core and iAB RBC 283 datasets, returning 10^3 times more molecular-versus-atomic EFMs.

By focusing on an atom of interest, our ACHMC approach greatly reduced the number of EFMs to biologically relevant pathways involving nutrient source remodelling. This reduction in pathways resulted in AEFM-versus-EFM enumeration completing $10^3 - 10^4$ times faster (Figure 3b). While running FluxModeCalculator in parallel did improve run times by 8.2-8.6 times (Figure S4), it remained over $10^1 - 10^2$ slower than enumerating the corresponding carbon and nitrogen AEFMs. One may think that expanding metabolites into metabolite-atom positions would increase the number of ACHMC states and therefore the computational run time. However, we found that AEFM enumeration actually involved fewer metabolite-atom positions than the total number of metabolites in a given network. Constrained by the atom mapping predictions, we observed that the average carbon and nitrogen ACHMC across all five GEMs only transited 0.41% and 1.7% of the total number of metabolite-carbon and metabolite-nitrogen positions, respectively (Figure S5). While serial run times in Figure 3c scaled quadratically (Figure 3c; $\text{slope} = 1.85, \text{intercept} = -6.54$), our AEFM enumeration approach benefits greatly from parallelism.

Not only are ACHMCs computed independently of one another, but our program is also parallelised on the level of individual ACHMCs to improve AEFM enumeration times. For the top 31 largest carbon ACHMCs in the HepG2 network, Figure 3d shows a median speedup of 7.9 times when using 16 or 32 threads.

Most AEFMs are source-to-sink pathways spanning dozens of reactions

We next turned to characterising the structural properties of all carbon and nitrogen AEFM across the five GEMs. Molecular EFMs can be divided into pathway that span source and sink metabolites or internal loops within the metabolic network such as pairs of reversible reactions. Both pathways have clear biological interpretations with source-to-sink pathways explaining the steady state flow of source metabolites on their way to exiting the network, and internally looped pathways corresponding to substrate remodelling/transport cycles. At an atomic level, these pathways can be subclassified further into AEFMs that transit sets of distinct metabolites or those that revisit the same metabolite, albeit in different atomic positions. An example source-to-sink AEFM with this property is a glucose-derived carbon AEFM remodelling through two turns of the TCA cycle; on the second turn, the position of the carbon atom shifts within the TCA intermediates, resulting in different isotopomers (Duan et al, 2022) and therefore a source-to-sink pathway with a metabolite revisitation under our nomenclature. The remodelling of fructose 6-phosphate to fructose 1,6-bisphosphate via phosphofructokinase and fructose bisphosphatase constitutes a cyclic carbon AEFM.

To obtain a broad understanding of the types of AEFMs within the five GEMs, we categorised them into source-to-sink pathways and looped pathways with or without metabolite revisitations in different atomic positions. Figure 4a-b shows the counts for each AEFM classes within each GEM. As expected, we observed that the majority of carbon AEFMs were source-to-sink pathways with or without metabolite revisitations. Those without metabolite

revisitations tended to scale as a function of network size, while those with revisitations increased modestly. Investigation of these source-to-sink pathways confirmed that many metabolite revisitations involved TCA cycle intermediates. Looped AEFMs without metabolite revisitations primarily corresponded to transport reactions cycling metabolites across compartments and cyclic pathways involving TCA intermediates. These two observations explain, in part, why the iAB RBC 283 network contained the fewest number of pathways with metabolite revisitations since they lack organelles and cannot support cyclic AEFMs associated with metabolite transport between compartments nor mitochondrial TCA activity. There were very few internally looped AEFMs with metabolite revisitations with no obvious trend across the five GEMs. The greatest number of these pathways were found in the E. coli core network with many metabolite revisitations involving carbon dioxide derived from isocitrate and alpha-ketoglutarate re-entering the TCA cycle via oxaloacetate. As there are fewer metabolites containing nitrogen, we found the majority of their AEFMs corresponded to pathways without metabolite revisitations, primarily involving amino acid remodelling or transports between compartments. Looped pathways with metabolite revisitations were only observed in the iIT341 and iSB619 networks through reactions involving the production and consumption of ammonia.

Since many looped AEFMs involve reversible reactions or metabolite transport across compartments, we hypothesised that looped AEFMs should involve fewer reactions than source-to-sink pathways which can span numerous internal metabolites. We calculated the fraction of carbon and nitrogen AEFMs with given lengths across all five GEMs and plotted their distributions in Figure 4c-d. Our results show that carbon AEFMs in the first four networks are bimodally distributed with the majority of pathways involving a dozen or

fewer reactions. While carbon AEFMs of the HepG2 network appeared to exhibit a single Gaussian distribution ($\mu = 67.7$, $\sigma^2 = 16.2$), closer inspection of the histogram revealed a small peak spanning 2-12 reactions. To determine whether these peaks were associated with source-to-sink and looped AEFM classes, we reclassified the carbon AEFMs spanning less than 5-12 reactions depending on the GEM (shorter pathways), and the remaining carbon AEFMs (longer pathways). We found that longer pathways were associated with source-to-sink pathways, with shorter pathways containing the majority of looped pathways. In fact, 65-97% of all longer pathways in each GEM were source-to-sink pathways with metabolite revisitations, with this statistic increasing to 95-100% when including source-to-sink pathways without metabolite revisitations. Among shorter pathways, we observed that roughly half or more of them were looped carbon AEFMs with or without metabolite revisitations. Analysis of the nitrogen AEFMs revealed only a single distribution of relatively short pathways, since there are much fewer nitrogen-containing metabolites across all GEMs.

We next investigated how atoms within the same source metabolite may follow different paths through the network, irrespective of their AEFM class. We first computed the number of AEFMs for each carbon atom (resp. nitrogen atom). For each metabolite, we then computed the ratio of the class. We then computed the number of AEFMs for each carbon atom (resp. nitrogen atom). So for example, a ratio of 10 means that one of the carbon atoms in the metabolite can transit the network in 10 times as many different ways as the carbon with the fewest paths through the network. Our results are shown in Figure 4e-f, where each dot corresponds to the ratio for a given source metabolite. We found that a surprising number of carbons and nitrogens can transit different pathways within their respective source metabolites. Focusing on the source metabolites with carbon and nitrogen

ratios over 100, we observed many of these corresponded to hydrophobic and positively-charged amino acids such as phenylalanine, methionine, arginine, and histidine. Source metabolites containing fewer carbons or nitrogens tended to exhibit ratios closer to 1 with AEFMs involving the same metabolite subsets. Metabolites such as glycine, for example, were equivalently metabolised across the same carbon AEFMs. For each GEM, and not counting source metabolites with ratios of one, we computed median carbon ratios between 2.0 and 41.8 and nitrogen ratios between 1.3 and 11.2. Overall, these findings highlight the varying degrees of metabolic flexibility for the atomic constituents of source metabolites to transit the metabolic network.

A minority of carbon AEFMs explains the majority of source metabolite remodelling

In the previous sections, we showed how to enumerate carbon and nitrogen AEFMs in large-scale networks and analyse the structure of these pathways. Yet, an equally important focus of our work is to determine the amount of flux moving through AEFM. This computational question is biologically motivated to identify the most important pathways among the thousands to millions of AEFMs. Furthermore, AEFMs with the greatest weight, and therefore carrying the most atomic flux, may correspond to existing metabolic subsystems or novel metabolic pathways. In the context of the HepG2 network, which contains a single set of steady state fluxes estimated during the exponential growth phase (Nilsson et al, 2020b), it is well-known that cellular metabolism depends on the non-essential amino acid glutamine for survival and proliferation (Ye et al, 2023; Yoo et al, 2020). Hence, a AEFM analysis of

glutamine carbons may reveal alternative paths of glutamine metabolism and quantify the amount of flux moving through these pathways.

Using our full AEFM pipeline, we decomposed the set of steady state fluxes onto the 448,629 glutamine-derived carbon AEFMs characterising the HepG2 metabolic network. Figure 5a shows the cumulative explained mass flow of each glutamine carbon as a function of their AEFMs ranked from greatest to smallest explained mass flow. We capped the cumulative explained mass flow of each glutamine carbon at 99% because we observed that 99.5% of the total AEFMs collectively explained the remaining 1% glutamine carbon flow. These results suggested that the majority of glutamine carbon remodelling is explained by a surprisingly small number of glutamine-derived carbon AEFMs, especially since our Markovian flux decomposition approach assigns some weight to all network pathways. The top 5 ranked AEFMs across each glutamine carbon (25 pathways altogether) explained nearly half (43.9%) of the total glutamine carbon mass flow with the subsequent 5 next highest ranked AEFMs only explaining an additional 8.1% glutamine carbon mass flow. These observations were consistent across carbon AEFMs derived from other amino acids as well as glucose, leading us to hypothesise that steady state pathway fluxes are centralised to a relatively small subset of AEFMs (Figure S6). For some source metabolites, we observed overlapping curves between carbons (e.g. arginine and glycine in Figure S6), suggesting that individual carbons were remodelled through the same AEFMs. However, corroborating results from Figure 4e-f, we observed many non-overlapping curves for the majority of amino acids since their carbons could transit the network via different internal metabolites.

To obtain a high-level understanding of the dominant pathways of glutamine carbon remodelling, we constructed a metabolic subnetwork from the top 5 glutamine carbon AEFMs,

collapsing metabolite-atom positions onto distinct metabolites, since none of these AEFMs involved metabolite revisitations. Of the 435 metabolites in the HepG2 dataset, input glutamine can remodel into 203 species localised to the cytosol or mitochondria. We found that the top 5 glutamine-derived carbon AEFMs consist of just 22 metabolites and 29 reactions (Figure 5b). These AEFMs generally consisted of cyclic pathways involving TCA intermediates. We found several source-to-sink AEFMs involving the production of carbon dioxide via the decarboxylation of isocitrate in the cytosol and alpha-ketoglutarate in the mitochondria.

As AEFMs are steady state pathways, they can be visualised as Sankey diagrams to better understand the atomic fluxes through the atomic flux subgraph. Figure 5c-e shows the top 5 AEFMs for each glutamine carbon. Each AEFM in the diagram is shown as a flow proportional to its AEFM weight with nodes representing the metabolite transited by the given glutamine carbon. Although each glutamine carbon can transit a different number of AEFMs, we found the top 5 backbone carbons for the first three sidechain carbons were identical in AEFM pathway with approximately the same weights.

Focusing on the first glutamine carbon in Figure 5c, we observed that AEFMs with the greatest weight corresponded to well-established metabolic subsystems. The first, third, and fourth AEFMs, for example, corresponded to aspects of the malate-aspartate shuttle. The fifth AEFM involved glutamine remodelling into alpha-ketoglutarate and entering the canonical TCA cycle and leaving as carbon dioxide via isocitrate. We also noticed that the second largest AEFM cycled carbon mass flow between TCA intermediates citrate, oxaloacetate, and malate between the cytosol and mitochondria. Interestingly, this metabolic pathway corresponds to the recently discovered non-canonical TCA activity described by Arnold et al, who validated this pathway in non-small cell lung cancer, mouse embryonic,

and C2C12 mouse myoblast cells by stable isotope tracing analysis (Arnold et al, 2022). This non-canonical TCA activity was further observed in glutamine carbons 2-4 in the second AEFM (Figure 5d). We also found first, third, fourth, and fifth AEFMs in in Figure 5d-e corresponded to the malate-asparate shuttle. The terminal glutamine carbon in Figure 5e contained three source-to-sink AEFMs that either transported glutamine out of the system or decarboxylated it into carbon dioxide via alpha-ketoglutarate.

Thus far, we have shown that our analyses revealed that the most active glutamine-derived carbon AEFM weights corresponded to well-known metabolic subsystems. This finding led us to ask whether we could summarise our AEFM analysis as a simplified metabolic model describing over 40% of the glutamine carbon remodelling in the HepG2 flux network. We present the resulting schematic in Figure 5f which consists of the following three major metabolic pathways: canonical TCA cycle, non-canonical TCA cycle, and the malate-aspartate shuttle. The overall schematic suggests that input glutamine is primarily routed into the mitochondria via aspartate/glutamate antiporters encoded by the *SLC25A12* and *SLC25A13* annotated in the HepG2 metabolic model. This mitochondrial glutamate is used to regenerate TCA cycle intermediates by converting glutamate into alpha-ketoglutarate which is subsequently consumed in the canonical TCA cycle to produce carbon dioxide and NADH/ATP/$FADH_2$. More importantly, a greater proportion of glutamate appears to be recycled through the non-canonical TCA cycle, which does not generate NADH, but actively transports existing NADH from the cytosol into the mitochondrial. Taken together, the dominant carbon AEFMs predicted by our model suggest HepG2 glutamine both fuels TCA cycle activity and regulates cellular levels of NAD^+/NADH.

Discussion

Since the proposal of molecular EFMs, much effort has been made to enumerate and decompose steady state metabolic fluxes onto these pathways. By defining EFMs with respect to individual atoms rather than molecules, we introduced the AEFM which describes how atomic constituents are remodelled within a metabolic flux network. As AEFMs never involve branching pathways, they may be easier to interpret than their molecular EFM counterparts, which may involve multiple inputs and outputs to balance molecular stoichiometries. Using the atom mapping program RXNMapper, we adapted our previously described CHMC method to efficiently enumerate carbon and nitrogen AEFMs in five GEMs and characterise source metabolite remodelling at the atomic level. By comparing the number of AEFMs across atoms within the same source metabolite, we revealed how certain amino acids displayed remarkable variability in the number of pathways transited by their constituent atoms. This analysis, in particular, may be useful for designing non-uniformly labelled stable isotope tracers to identify fluxes within specific metabolic subsystems. Our final AEFM weight analysis of the HepG2 network revealed that the majority of glutamine carbons were remodelled via well-known metabolic pathways such as the canonical TCA cycle but also novel pathways such as non-canonical TCA activity which has not been previously described in HepG2 cells.

The problem of molecular EFM enumeration has historically limited their applicability to small-scale networks. Many algorithmic advances have incrementally improved molecular EFM enumeration, yet never feasibly beyond networks with > 100 metabolites and reactions (Kamp & Schuster, 2006; Terzer & Stelling, 2008; van Klinken & Willems van

Dijk, 2016). Molecular EFM enumeration is generally regarded as infeasible for large-scale networks—so much so that numerous methods have been developed to enumerate a subset of molecular EFMs constrained by pathway length (de Figueiredo et al, 2009a), reaction thermodynamics (Gerstl et al, 2015), participating metabolites (Pey & Planes, 2014), participating reactions (Kaleta et al, 2009), gene regulatory rules (Jungreuthmayer et al, 2013), metabolic subsystems (Schuster et al, 2002; Schwartz et al, 2007), extracellular flux measurements (Soons et al, 2011; Jungers et al, 2011), or even random sampling (Machado et al, 2012). While our HepG2 results did show the majority of source metabolite fluxes were explained by a small number of relatively short AEFMs, contextualising the significance of these dominant pathway fluxes is only possible when all AEFMs are known. Furthermore, reaction thermodynamics, gene regulatory rules, and other constraints used to identify these EFM subsets may change across different biological conditions. By computing AEFMs instead, we could feasibly enumerate these pathways in large-scale networks containing hundreds of metabolites and reactions within minutes to hours on consumer-grade computing hardware. We are hopeful that algorithmic tricks borrowed from molecular EFM enumeration tools, such as linear pathway compression, could further decrease memory usage and run times.

Aside from carbon and nitrogen AEFMs, our work is broadly applicable to other types of chemical elements. One could feasibly enumerate oxygen or hydrogen AEFMs in small-scale networks, although the significance of these pathways is unclear for biochemical networks given the ubiquitous nature of these atoms. We also caution against enumerating hydrogen AEFMs, in particular, given limitations associated with RXNMapper. Unless explicitly represented in the reaction SMILES string, RXNMapper does not assign hydrogen atom mappings. Moreover, RXNMapper cannot map atoms in reaction SMILES strings longer than

512 characters. Our method could equally be applied to study sulphur metabolism to quantify methionine or cysteine remodelling, or phosphorus metabolism to quantify nucleotide synthesis. Outside the realm of biochemistry, our method is applicable to any periodic table element and possibly relevant for studying general chemical reaction networks.

Our AEFM framework opens many new computational problems in metabolism. While we analysed one set of fluxes for the HepG2 network, one could envision a differential AEFM analysis of flux distributions across distinct conditions. These differential AEFMs may reveal changes in metabolic subsystem activity or predict new pathways of source metabolite remodelling. For instance, analysing carbon AEFMs of C1 compounds may be prove useful for quantifying or optimising carbon fixation pathways in C1-utilising organisms. Another important question is whether a joint analysis of AEFM weights could reveal broader metabolic changes within a condition. We observed that many AEFMs across distinct source metabolites converged on common internal metabolites. Intuitively, the weights of AEFMs rooted on different source metabolites, but sharing common downstream metabolite subsequences, should be related, and these relationship may further speed AEFM enumeration or weight computations. Finally, more work is needed to establish relationships between AEFM decompositions and more traditional EFM decompositions of fluxes, and to explore the possibility that feasible AEFM calculations might somehow aid in molecular EFM calculations.

Materials and Methods

Methods and Protocols

The following subsections describe in greater detail the datasets and pipeline steps referenced in Figure 2.

Step 1a: Datasets

Five highly-curated GEMs were chosen in this study to reflect a diversity of organisms, metabolites and biochemical reactions. Four networks were obtained from the BiGG UCSD database (King et al, 2016) in ascending order of network size. Datasets were downloaded as SBML (.xml) files, and the corresponding stoichiometry matrices, metabolites and reactions were extracted using the Julia SBML.jl package. The HepG2 liver cancer cell line dataset was chosen for its single set of estimated steady state fluxes estimated from a FBA with external fluxes constrained by time-resolved exometabolomic measurements of glucose, pyruvate, lactate, and various amino acids. These cells were grown in DMEM with 10% fetal calf serum, 22mM glucose, and 1.8mM glutamine (Nilsson et al, 2020b). The steady state fluxes were computed from a piecewise FBA model provided by the authors (figure3_metabolitebalances.m; MATLAB R2018a) (Nilsson et al 2020b; Data ref: Nilsson et al 2020a). Across all models, we modelled all reversible reactions as net forward reactions to simplify the number of molecular and AEFMs as the majority of experimental and computational methods can only estimate net fluxes. We assumed the reaction stoichiometries of the BiGG datasets represented the canonical net directions of all reversible reaction. For the HepG2 dataset, the net directions were chosen based on the net fluxes estimated from

the FBA model.

Step 1b: Extracting metabolites SMILES strings

SMILES structures were not encoded in the BiGG SBML files nor the HepG2 dataset. Thus, we manually extracted SMILES strings corresponding to each metabolite across all GEMs. Isomeric SMILES strings were taken from PubChem (Kim et al, 2023), the Human Metabolome Database (Wishart et al, 2022), or MetaNetX database (Moretti et al, 2021). Metabolites shared across GEMs were assigned the same SMILES structure in addition to identical metabolites stratified across distinct subcellular compartments. Generic structures such as R-groups were not modelled in this study and were replaced with a single bonded hydrogen under the assumption this modification would not alter atom mappings, especially since RXNMapper does not map hydrogen atoms unless they are explicitly represented in the SMILES strings. SMILES strings were also not assigned to metabolites with ambiguous structures nor pseudometabolites with no known chemical structure. All pseudometabolites and reactions involving pseudometabolites were removed from the stoichiometry matrix (Table 2).

Step 1c: Constructing reaction SMILES strings

RXNMapper performs atom mapping on a string that encodes the SMILES structures of all substrates and products for a given reaction. These reaction SMILES strings are constructed by concatenating SMILES strings of individual substrates and products. Multiple substrates or products are separated by the symbol '.', while the reaction arrow delimiting substrates and products is denoted by the symbol '>>'. The substrate and product SMILES strings

are concatenated in order of metabolite appearance within the stoichiometry matrix, and multiple stoichiometric copies are consecutively repeated in the reaction SMILES strings. Non-integer stoichiometric coefficients are therefore not allowed and these pseudoreactions were removed from the stoichiometry matrix (see Table 2). To maintain steady state flux in the HepG2 network, flux along pseudoreactions were replaced with unimolecular source and sink fluxes consuming each pseudoreaction substrate and producing each pseudoreaction product.

Step 1d: Pre-processing genome-scale metabolic models

Several pre-processing steps were required to compute AEFMs from the GEMs. The stoichiometric coefficients of external reactions were set to one and their fluxes rescaled to carry equivalent units of source and sink metabolites. All reactions containing pseudometabolites with no known SMILES strings were removed from the stoichiometry matrix and replaced with unimolecular source and sink reactions carrying equivalent units of flux. The same procedure was also applied to reactions containing non-integer stoichiometries. These pseudometabolites were next removed from the stoichiometry matrix rows, and unimolecular reactions involving the same external or internal metabolites were aggregated. Finally, we had to remove a small number of reactions whose reaction SMILES strings exceeded the 512 token limit of RXNMapper. These reactions were likewise converted into unimolecular source and sink reactions to maintain steady state flux across the network (see Table 2). All pre-processing steps described above are wrapped into convenience functions within our Julia package MarkovWeightedEFMs.jl.

Step 1e: running RXNMapper to map atomic movements across reactions

After GEM pre-processing, RXNMapper was used to predict the atom mappings across reaction SMILES strings. The output of RXNMapper is a mapped reaction SMILES string with numeric indices indicating the atomic movements from substrate to product side of each reaction SMILES string. We manually checked several reactions to ensure the assigned atom mappings were correct. Crucially, we found RXNMapper to fail on nearly all classes of ATP-independent transaminase reactions. This problem has previously been reported in an atom mapping benchmark study prior to the publication of RXNMapper (Preciat Gonzalez et al, 2017). These transaminase reactions were identified in all metabolic networks and manually atom mapped based on the KEGG reaction mappings (Figure S7). Additionally, the reactions converting citrate to isocitrate (aconitate hydratase) and alpha-ketoglutarate to succinyl-CoA required manual curation.

We then store these curated atom mapping predictions in a GEM and atom type-specific atom mapping dictionary. Each key corresponds to a substrate-atom uniquely defined by its (i) metabolite index within the stoichiometry matrix, (ii) ordinal occurrence of that atom type within the substrate SMILES string, (iii) substrate stoichiometric copy, and (iv) reaction index within the stoichiometry matrix. The values of this dictionary are the corresponding product-atom positions uniquely defined by the (i) metabolite index within the stoichiometry matrix and (ii) ordinal occurrence of that atom type in the product SMILES string.

Step 2: Constructing atomic flux subgraphs

From the atom-mapping dictionary, we then construct atomic flux subgraphs rooted on each source metabolite-atom position. Starting with the source metabolite-atom position, we search the atom-mapping dictionary to determine the participating reactions and product metabolite-atom positions. For reactions involving a single stoichiometric copy of each substrate and product, the atom of interest will always transition from a unique substrate metabolite-atom position to product metabolite-atom position. If this atom is present in a reaction with multiple substrate stoichiometric copies, we assume there is an equal probability of that atom occurring in either substrate stoichiometric copy. Depending on the substrate stoichiometric position, the atom may (i) transition to different product metabolite-atom positions, (ii) the same product metabolite but at different atomic positions, or (iii) the same product metabolite-atom position. In all cases, we set the atomic flux of the atom transitioning from either stoichiometric copy equal to the reaction fluxes to maintain atomic mass conservation (Figure S1). The resulting atomic flux graph is completed by connecting any sink fluxes to the root of the tree initialised on the source metabolite-atom position. This final procedure ensures a closed-loop subgraph maintaining atomic mass conservation.

Step 3: Building the ACHMC model and enumerating AEFMs

The atomic flux subgraphs described in the previous step satisfy all conditions required to construct the corresponding ACHMC. Specifically, these conditions are that the input flux network is a closed-loop graph that is strongly connected with conserved edge weights such that the sum of weights entering and exiting each node are balanced (Chitpin & Perkins,

2023).

Rather than explicitly constructing an atomic flux subgraph for each source metabolite-atom position and the corresponding ACHMC, we choose to construct the ACHMCs directly from the metabolic flux network and the atom mapping dictionary. Starting at an initial metabolite-atom position, we apply a depth-first search algorithm (Algorithm 1) to recursively enumerate sequences of transited metabolite-atom positions stemming from the initial source metabolite atom. Following Chitpin & Perkins, we denote state S in the ACHMC to represent the simple path of metabolite-atom transitions in the trie $s_0, s_1, \ldots, s_m$. Linear AEFMs (simple paths) are enumerated by finding all states S (sequences of metabolite-atom positions) that can no longer transition to a longer simple path (leaves within the trie). Internally-looped AEFMs (simple cycles) are enumerated by finding all prefixes that are capable of transitioning to a shorter simple path within the trie.

Algorithm 1 Depth-first construction of the ACHMC trie

Input: Input state s_{im} denoting an atom in position i of metabolite m, dictionary D mapping $\{s_{im}\}_r$ to $\{s_{jn}\}_r$ in reaction r, stoichiometry matrix S, steady state reaction flux vector v

Output: Dictionary T encoding the ACHMC trie

```
function TRAVERSE_TRIE(s_im, D, S, v)
    V ← {s_im} Array of visited metabolite-atom positions
    W ← V[end]
    T ← {}  where T[Prefix] ← {Children, Children values, Parent, Parent values}
    R ← reaction indices in S where W participates as a substrate
    s_out ← {D[W]}_{r∈R}
    s_down ← s_out ∉ V
    s_up ← s_out ∈ V
    Add {s_down, v[r ∈ R], s_up, v[r ∉ R]} to T[V]
    for s ∈ s_down do
        TRAVERSE_TRIE({V, s}, D, S, v)
```

Step 4: Computing AEFM weights in the HepG2 network

To compute AEFM weights, we encode the ACHMC trie into a sparse transition probability matrix T. Each element $T_{S,S'}$ corresponds to the probability of an atom in state $S \equiv s_0, s_1, \ldots, s_m$ transitioning to either state $S' \equiv s_0, s_1, \ldots, s_m, s_{m+1}$ or state $S' \equiv s_0, s_1, \ldots, s_i$ if s_{m+1} is among $s_0, s_1, \ldots, s_m$. If no such state s_{m+1} exists, we set $T_{S,S^0} = 1$ to ensure the ACHMC is strongly connected where $S^0 \equiv s_0$. The transition probabilities are taken from the atomic flux values stored within the trie and normalised such that $\sum_{S' \notin S} T_{S,S'} = 1$.

The AEFM probabilities are computed by steady state analysis of the ACHMC transition probability matrix (Chitpin & Perkins, 2023). For AEFM k, we identify all transitions in the ACHMC that close the simple cycle E_k. For linear AEFMs, E_k always corresponds to one transition; however, internally looped AEFMs may contain multiple transitions within E_k. The steady state probability of transiting AEFM k is defined as the sum of steady state probabilities of those transitions or:

$$P_k = \sum_{(S,S') \in E_k} \pi_S \cdot T_{S,S'} \tag{1}$$

These AEFM probabilities are scaled to weights under the conservation constraint that the atomic flux v_0' entering a given source metabolite is equal to the observed (unimolecular) reaction flux v_0 of the underlying metabolic flux network. Thus for each ACHMC, we identify all l linear AEFMs and compute the ACHMC-specific scaling constant α satisfying $v_0 = \sum_{k \in l} P_k \cdot \alpha$.

Benchmarking the enumeration of atomic-versus-standard EFMs

We benchmarked our Julia package MarkovWeightedEFMs.jl version 2.0 running Julia version 1.10.2 to enumerate AEFMs versus the molecular EFM enumeration program FluxModeCalculator (van Klinken & Willems van Dijk (2016)) running MATLAB version R2018a with default parameters including network compression. We chose FluxModeCalculator because it is the state-of-the-art program for enumerating molecular EFMs. Benchmarks were performed on a 64-bit Linux operating system with 64GB of memory and an AMD 5950X CPU with 16 cores/32 threads. All benchmarks reported the wall time to run the AEFM or molecular EFM enumeration functions and did not include startup times to open Julia/MATLAB, precompile Julia functions, or load packages and input data. A benchmark time of 'did not finish' (DNF) was assigned if either program could not complete (A)EFM enumeration within 7 days. In practice, only FluxModeCalculator failed to enumerate the molecular EFMs in three of the five datasets.

Data availability

Our software package MarkovWeightedEFMs.jl is available at `https://github.com/jchitpin/MarkovWeightedEFMs.jl`. All source codes to reproduce this study are available at `https://www.github.com/jchitpin/reproduce-efm-paper-2024`.

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Author contributions

Justin G. Chitpin: Conceptualisation; data curation; methodology; software; formal analysis; investigation; visualisation; writing - original draft; writing - review and editing.

Theodore J. Perkins: Conceptualisation; methodology; supervision; funding acquisition; investigation; project administration; writing - review and editing.

Disclosure and competing interests statement

The authors declare no competing interests.

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Table 1: Summary statistics for the five metabolic networks in this study.

	Carbon metabolism				
GEM (# mets | # rxns)	Sources	Sinks	States	Transitions	AEFMs
E. coli core (71 | 144)	184	255	14659	24355	6519
iAB_RBC_283 (274 | 465)	954	1166	37122	51220	7958
iIT341 (433 | 759)	1486	2134	86342	120545	30633
iSB619 (547 | 974)	2188	2986	169877	246712	59444
HepG2 (435 | 545)	376	939	14159860	18152623	2180613
	Nitrogen metabolism				
GEM (# mets | # rxns)	Sources	Sinks	States	Transitions	AEFMs
E. coli core (71 | 144)	30	40	246	448	204
iAB_RBC_283 (274 | 465)	161	214	6409	9275	2091
iIT341 (433 | 759)	314	447	27440	39761	10440
iSB619 (547 | 974)	353	516	36661	54165	14214
HepG2 (435 | 545)	72	190	19137	27860	5105

Table 2: Number of metabolites and reactions removed from each network during preprocessing.

	E. coli core	iAB_RBC_283	iIT341	iSB619	HepG2
Rxns with non-integer stoichiometries	2	1	3	34	8
Pseudometabolites with no known structures	1	68	52	108	33
Rxns with pseudometabolites	2	142	93	204	46
Rxns exceeding RXNMapper token limit	0	1	4	14	3

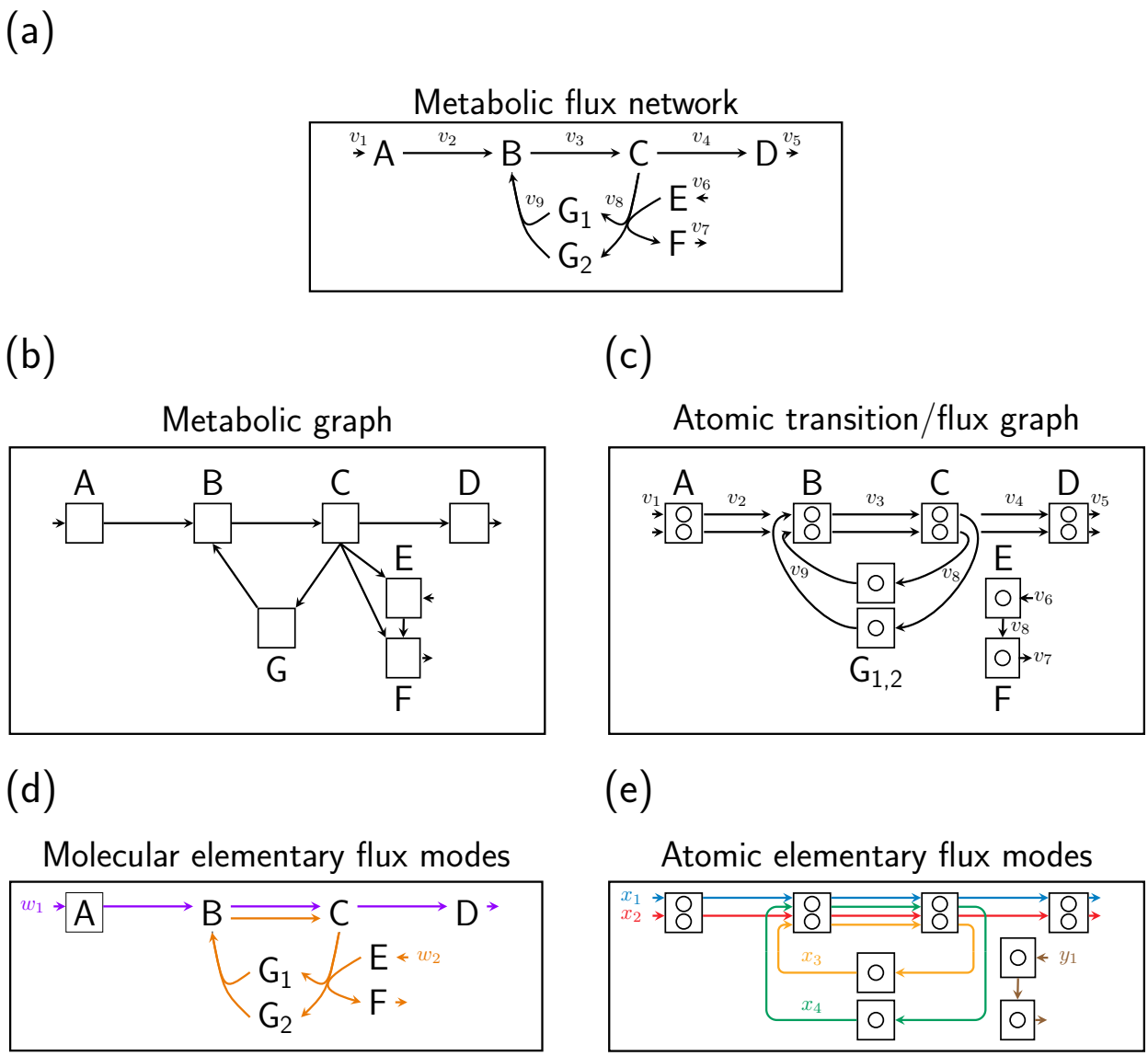

Figure 1: Various notions of metabolic networks with respect to an (a) example metabolic flux network, (b) metabolic graph, and (c) atomic transition/flux graph. The corresponding elementary flux modes are shown for the (d) metabolic flux network and (e) atomic transition/flux graph.

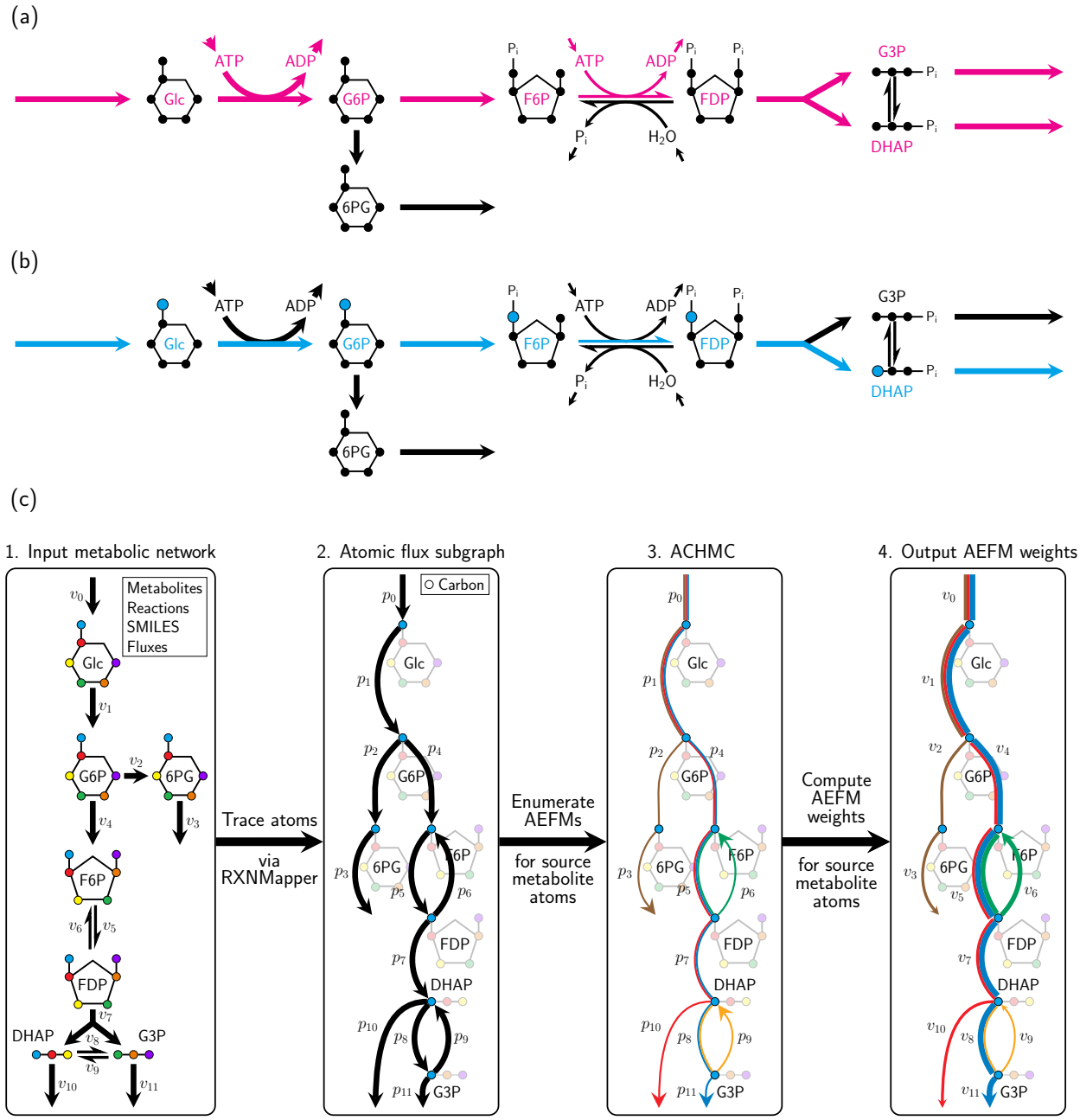

Figure 2: Atomic cycle-history Markov chain (ACHMC) pipeline. (a) An example molecular EFM in glycolysis. (b) An example AEFM tracing the flow of a given glucose carbon in glycolysis. (c) The major steps within our pipeline to enumerate AEFMs and compute their weights.

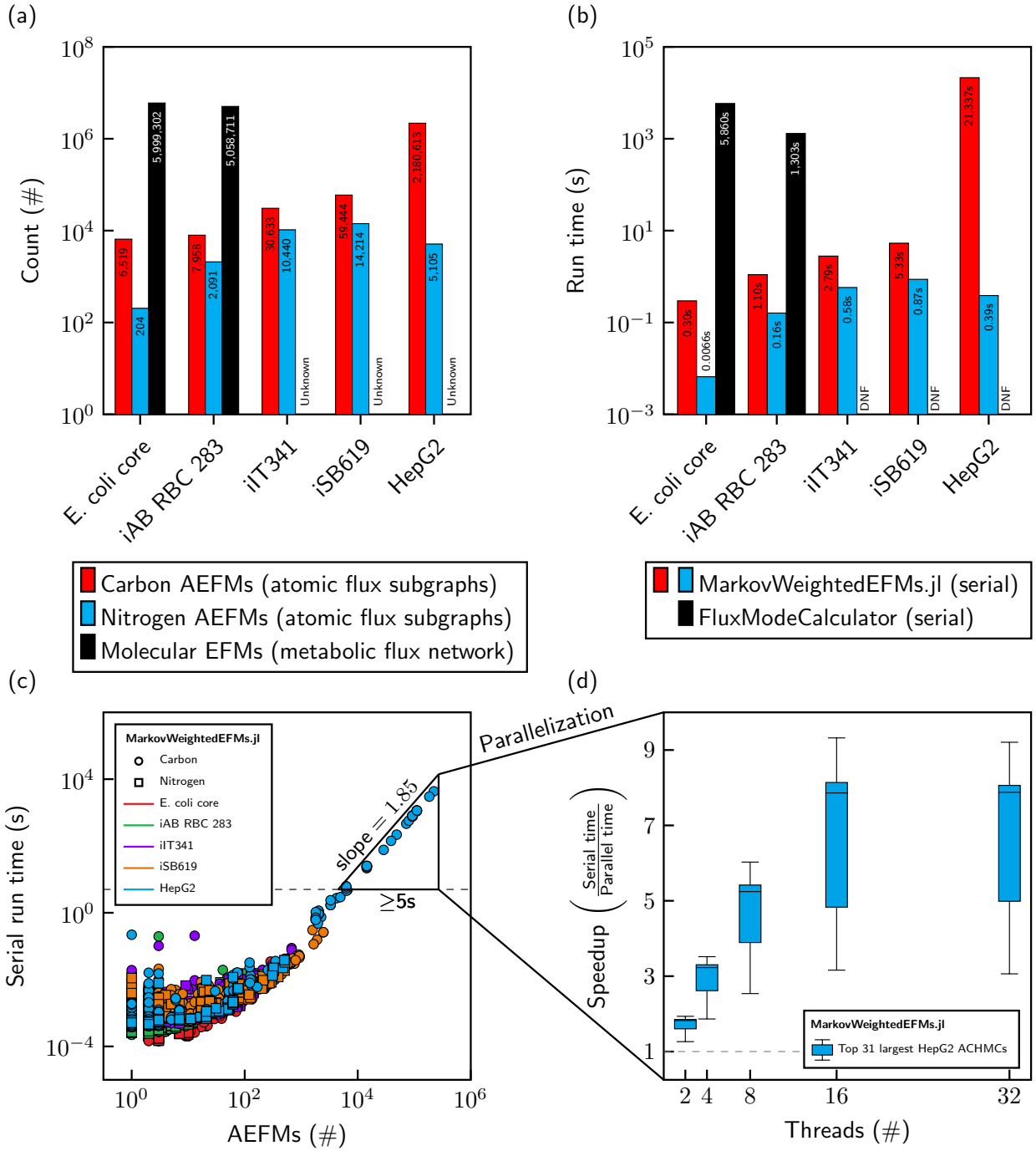

Figure 3: AEFM enumeration is computationally tractable compared to molecular EFMs in five GEMs. (a) Number of atomic and molecular EFMs in the five GEMs. (b) The run time of our MarkovWeightedEFMs.jl method versus FluxModeCalculator running in serial. (c) Serial run time scaling of MarkovWeightedEFMs.jl as a function of the number of AEFMs. (d) Speedup of selected ACHMCs as a function of additional threads.

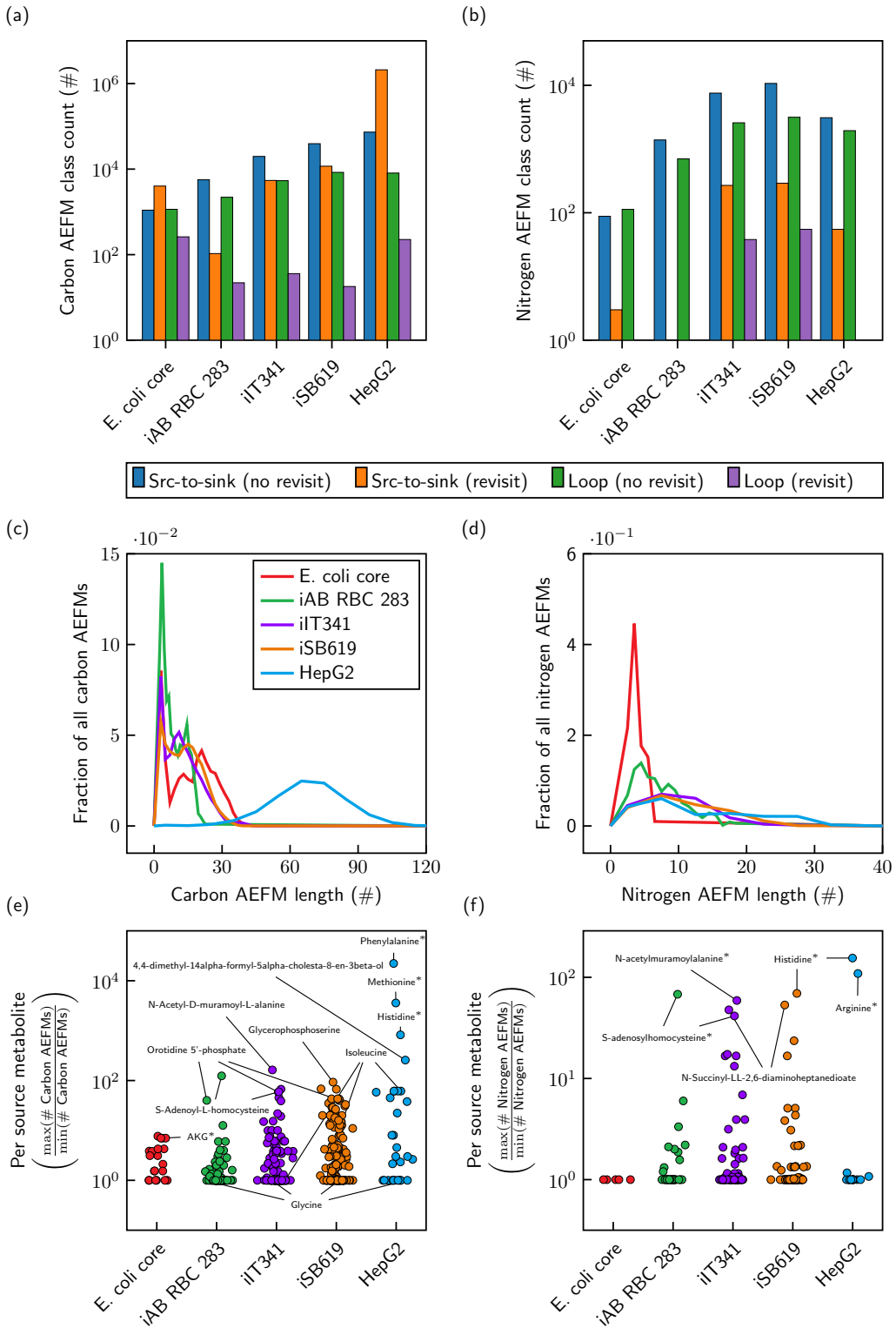

Figure 4: Structural analysis of AEFMs in five genome-scale metabolic models. (a-b) Classes of carbon and nitrogen EFMs. (c-d) Fraction of carbon and nitrogen AEFM lengths. (e-f) Ratio of the greatest and smallest number of AEFMs between carbons and nitrogens within the same source metabolite. Representative metabolites from each dataset are labelled; labels with asterisks indicate that metabolite is also present in other networks.

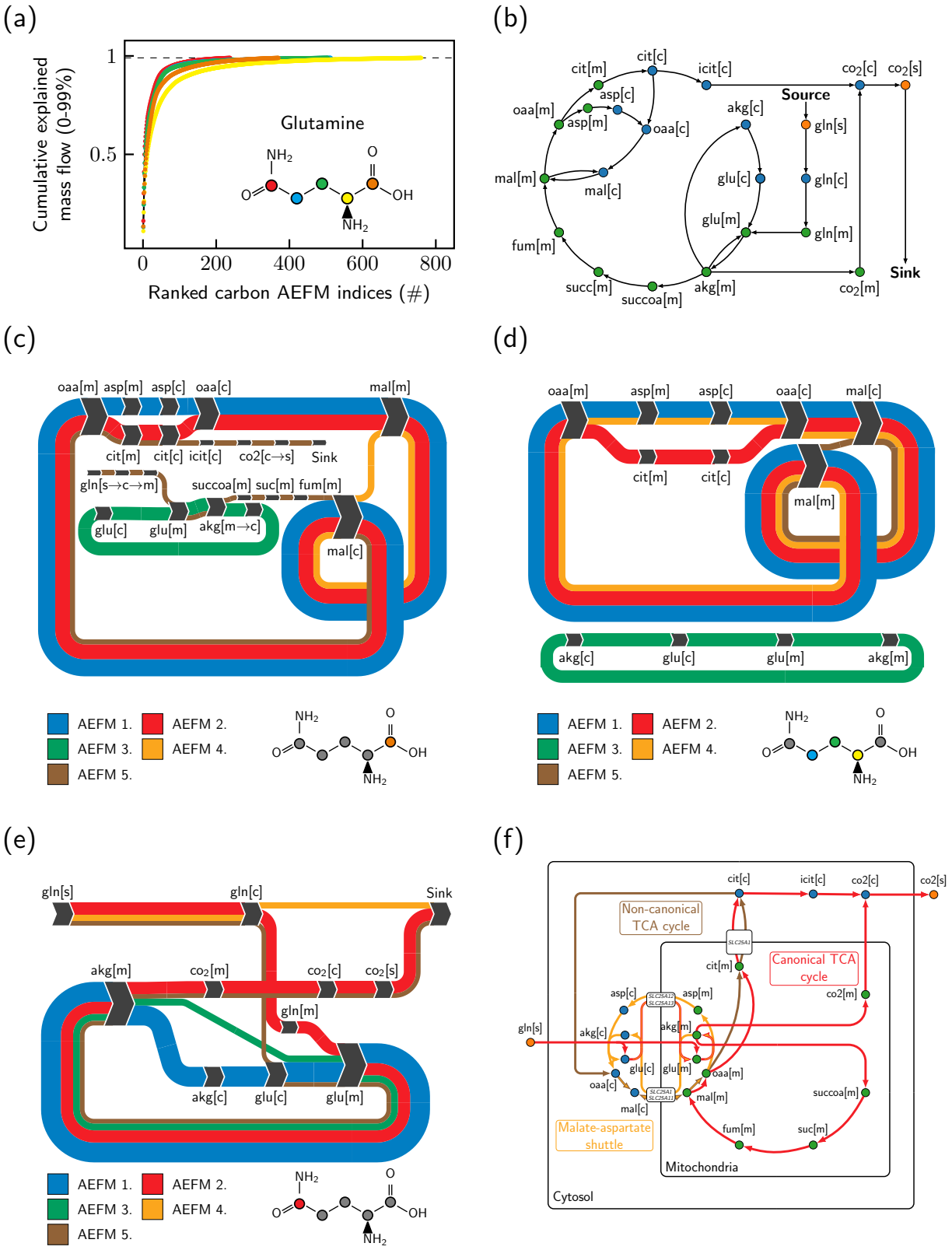

Figure 5: Carbon AEFM weight analysis of the HepG2 metabolic flux network. (a) A small number of highly active AEFMs explain the majority of glutamine carbon flow. (b) Subnetwork of the top five AEFMs across each glutamine carbon collapsed onto metabolites. (c-e) Sankey diagrams of the top five glutamine carbon AEFMs. (f) Schematic mapping AEFMs onto traditionally defined metabolic pathways/subsystems. All metabolite names are labelled following the BiGG database nomenclature.

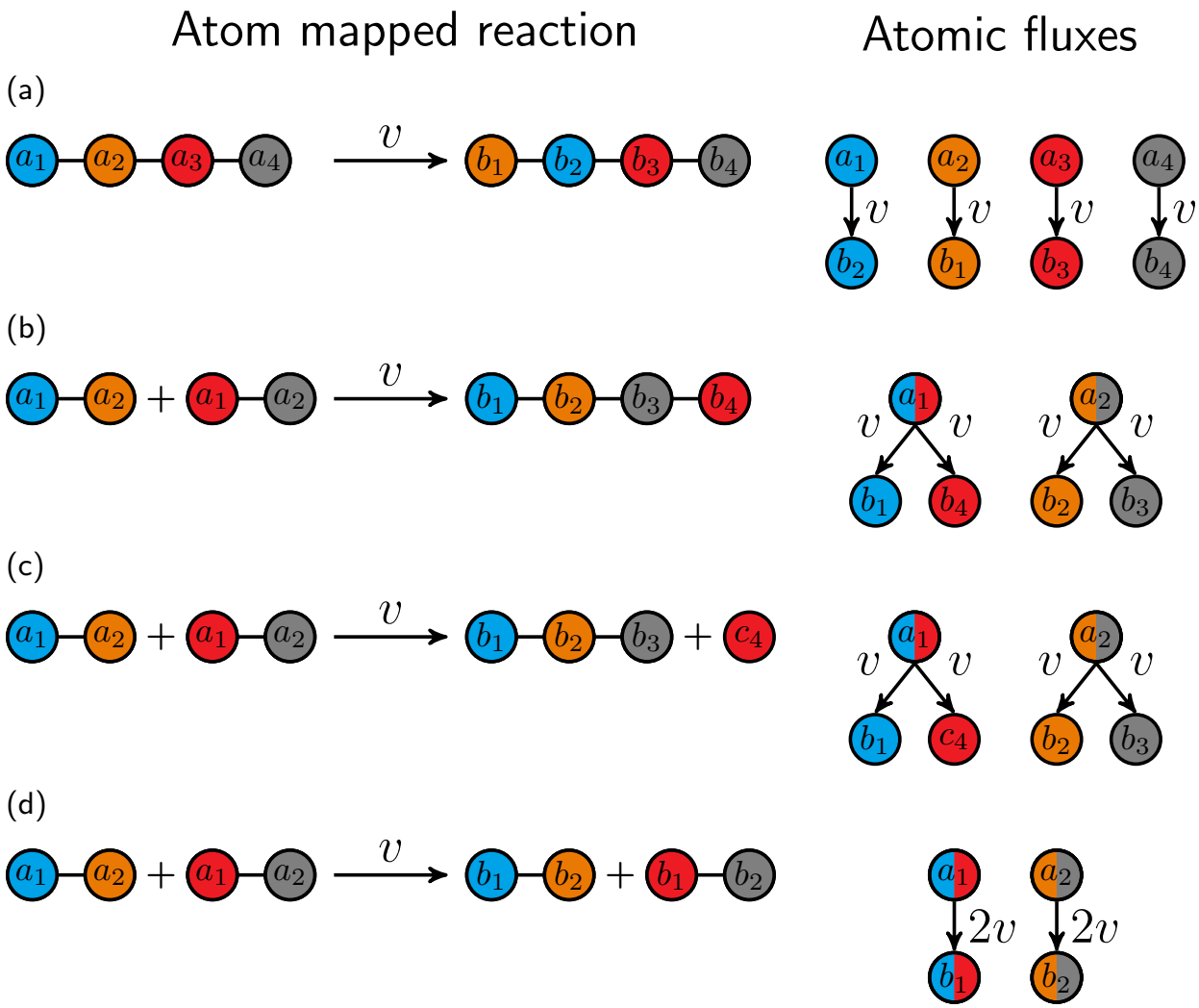

Figure S1: Rules for assigning atomic fluxes from atom-mapped reactions with biological examples. Schematic node colours represent ground truth atom mappings across substrate(s) and product(s). Node labels correspond to distinct atomic indices of substrate and product stoichiometric copies. (a) For reactions with single stoichiometric copies of substrates and products, each atom within a given substrate moves to a single atomic position within a product with atomic flux equal to the reaction flux. (b-d) For reactions with s stoichiometric copies of a given substrate, there are s copies of each metabolite-atom state on the substrate and product side of the reaction. The probability of any atom within a substrate stoichiometric copy moving to a product metabolite-atom state remains equal to the reaction flux. (b) The stoichiometric copies of the substrate fuse together into a single product. (c) The stoichiometric copies of the substrate form distinct products. (d) The s stoichiometric copies of the substrate form s copies of products with identical atomic backbones.

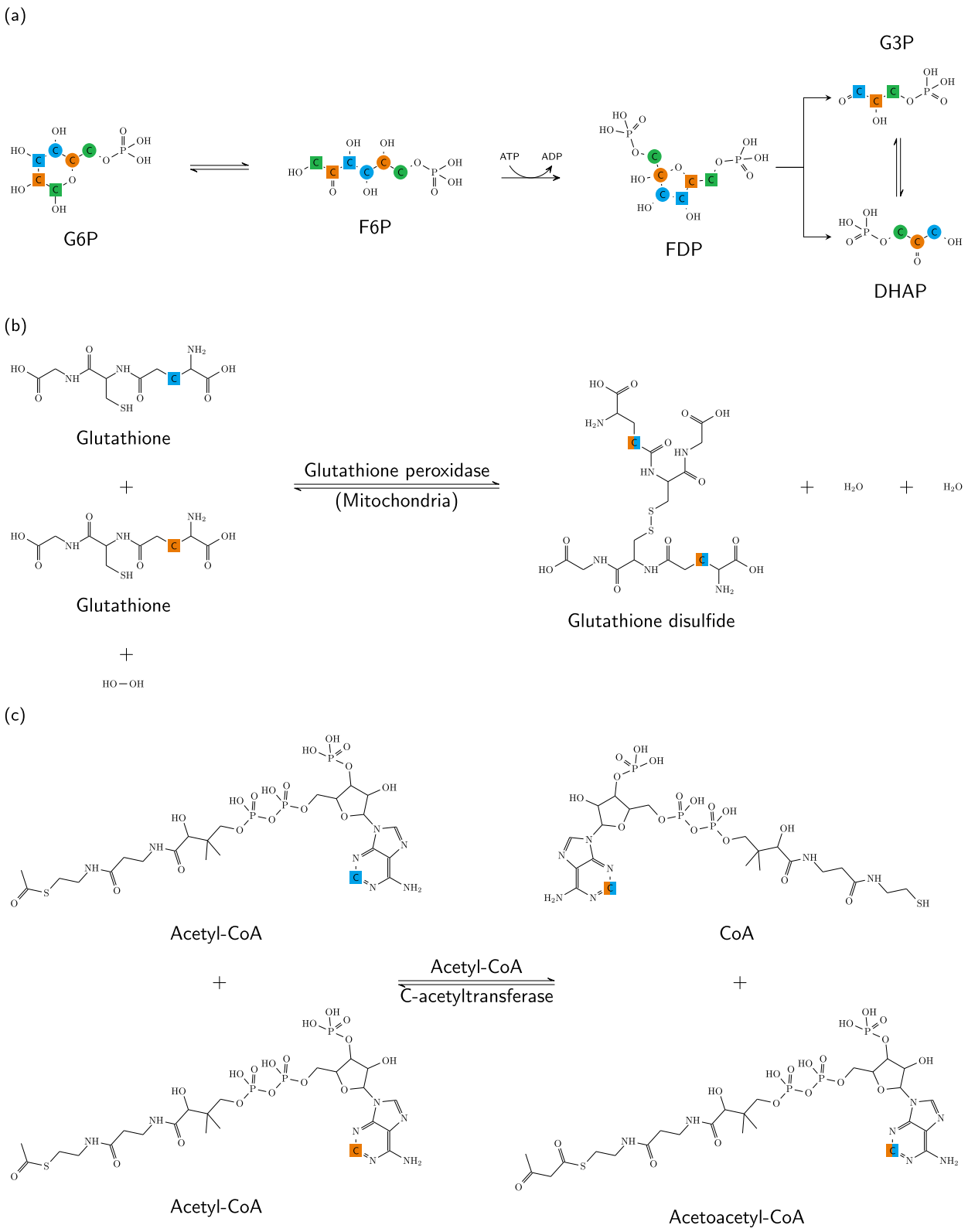

Figure S2: Biological examples for rules in Figure S1. (a) Biological example of rule a where the atoms in each substrate map to a single position on the corresponding product. (b) Biological example of rule b where two stoichiometric copies of glutathione fuse together to form a single stoichiometric copy of glutathione disulfide. (c) Biological example of rule c where two stoichiometric copies of acetyl-CoA undergo a reaction to produce a stoichiometric copy of two distinct products.

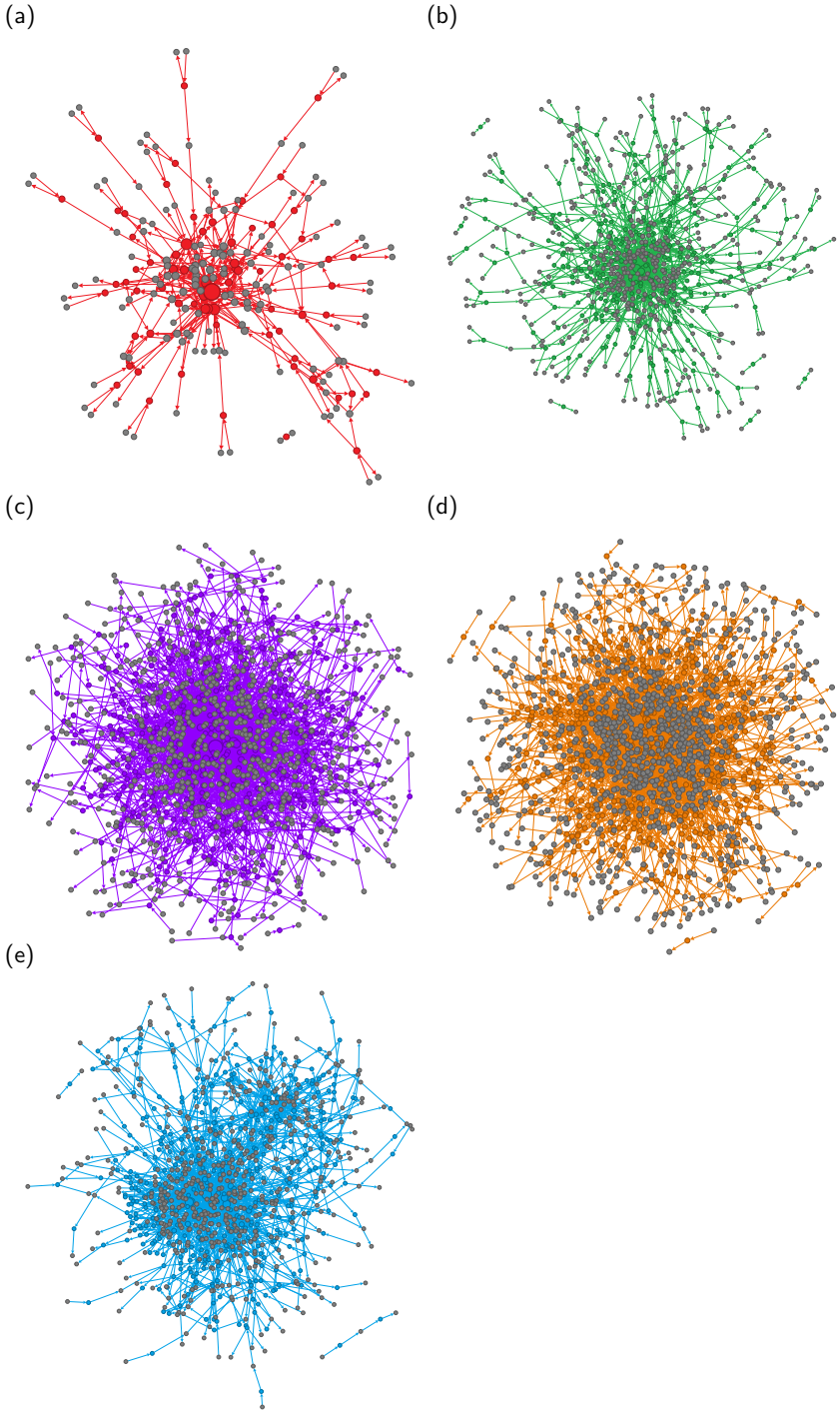

Figure S3: Network visualisation of the five GEMs after pre-processing. Metabolites (non-grey nodes) are connected to reaction entities (grey nodes) with node sizes proportional to their degree. Graphs were constructed in Gephi version 0.10 using the Yifan-Hu proportional network layout with default parameters. (a) E. coli core. (b) iAB RBC 283. (c) iIT341. (d) iSB619. (e) HepG2.

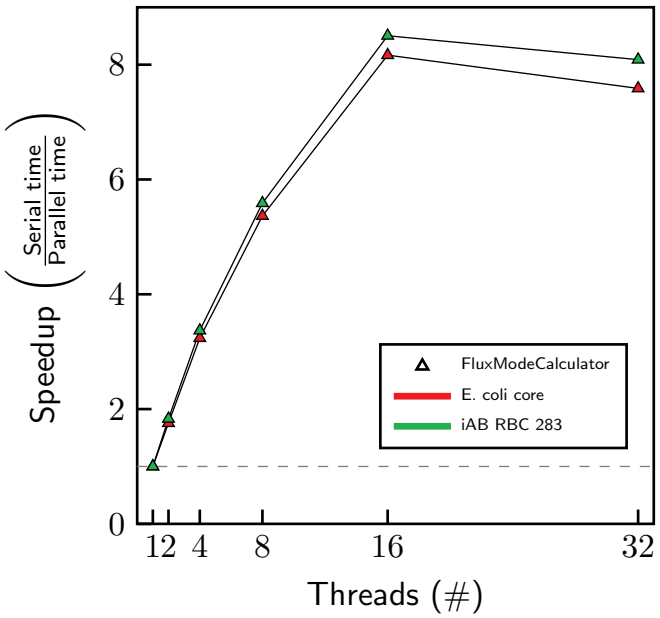

Figure S4: FluxModeCalculator scaling across multiple threads.

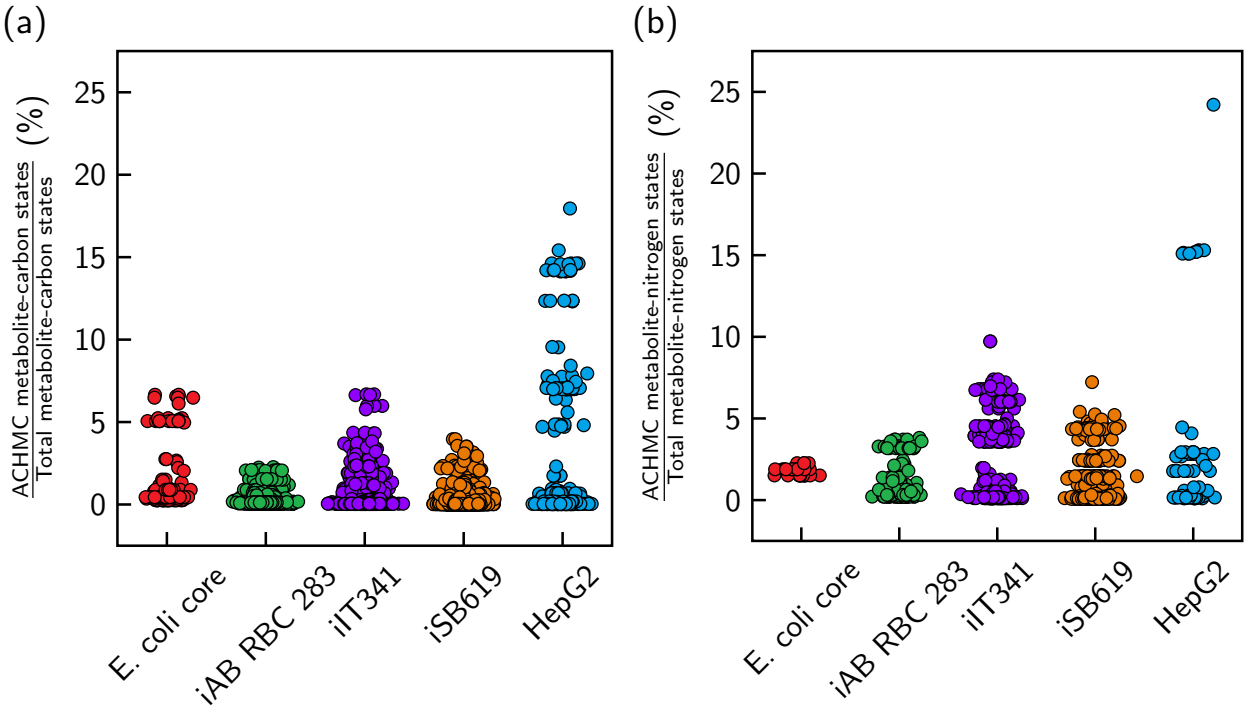

Figure S5: Most ACHMCs traverse less than 10% of the total metabolite-carbon/nitrogen state space. Each point corresponds to an ACHMC rooted on a given source metabolite carbon or nitrogen.

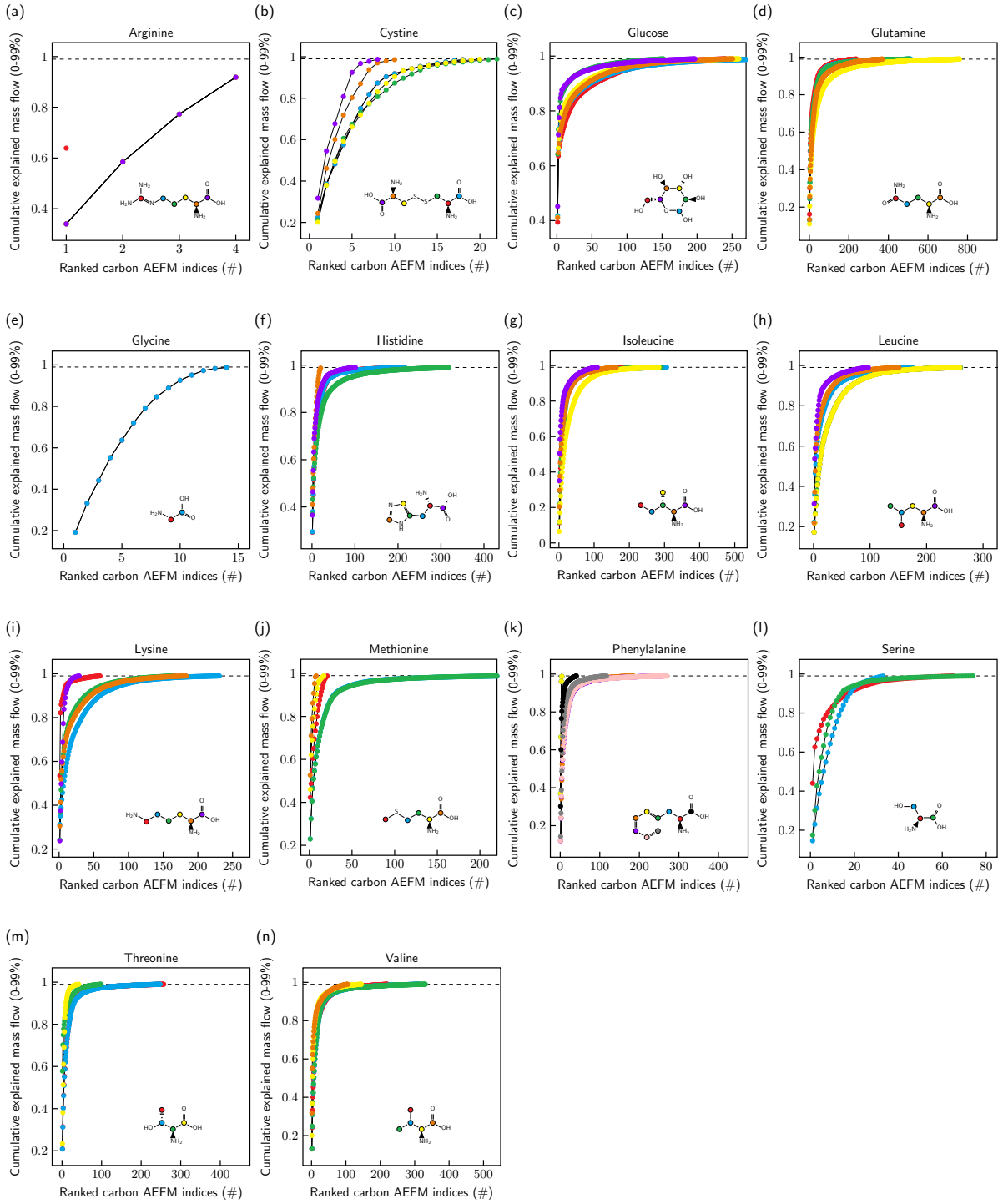

Figure S6: Cumulative explained carbon mass flow of input glucose and amino acid carbons from the HepG2 flux network. The cumulative explained mass flow is truncated at 99% to avoid displaying AEFMs that carry little atomic mass flow.

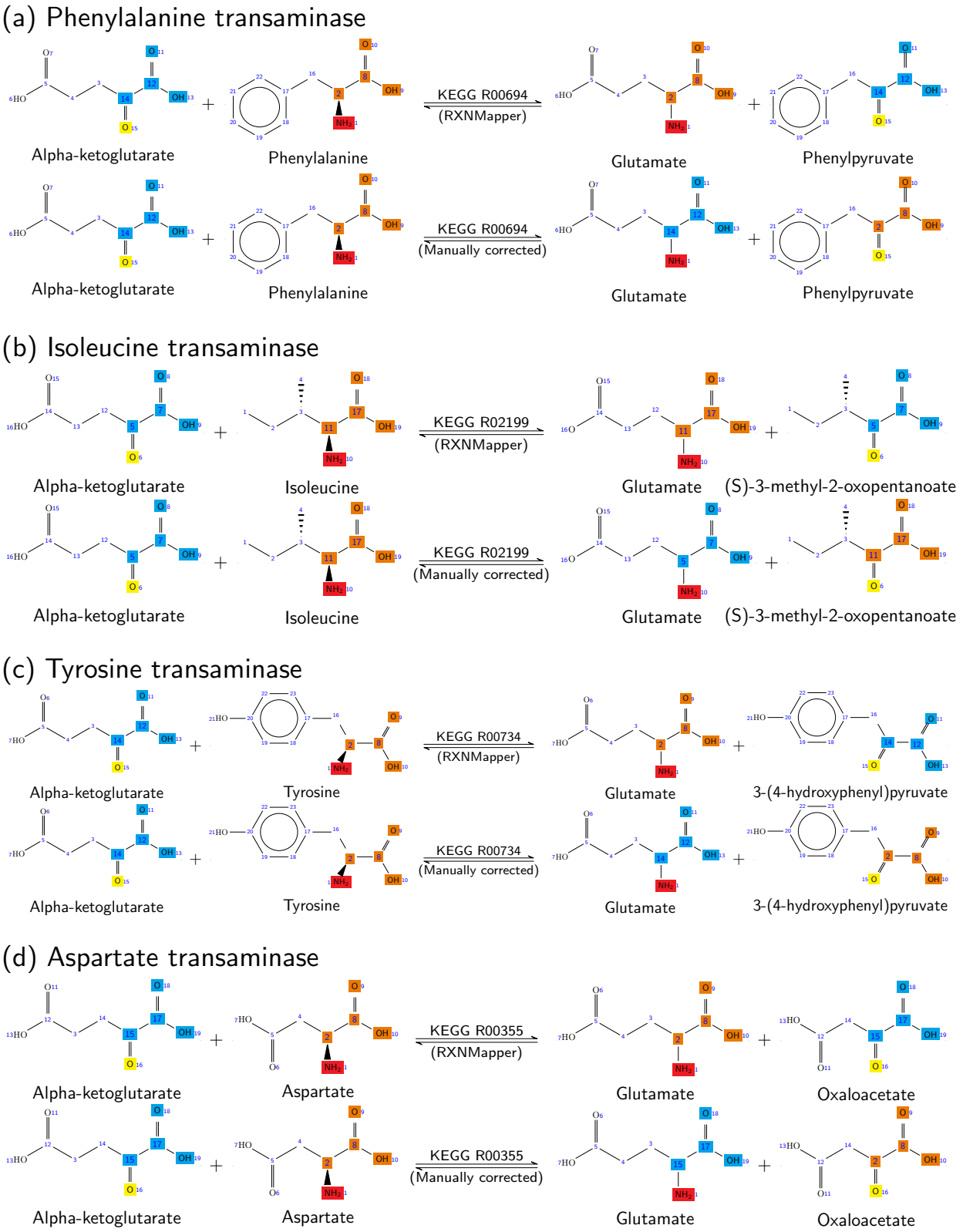

Figure S7: Examples of incorrectly atom-mapped transaminase reactions present in one or more of the five networks. (a) Phenylalanine transaminase. (b) Isoleucine transaminase. (c) Tyrosine transaminase. (d) Aspartate transaminase.